Hybrid Linear-Convex Confidence: An Asymmetric Scoring Function for Language Model Calibration

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Abstract

Language models are systematically overconfident, yet the standard tools for measuring and training calibrated confidence—Brier score, log-loss—treat over- and under-confidence symmetrically. We propose the Hybrid Linear-Convex Confidence (HLCC) scoring function, which pairs a linear reward $R(c) = 1+c$ for correct predictions with a quadratic penalty $P(c) = -2c^2$ for confident errors. The resulting optimal confidence $c^* = p/[4(1-p)]$ is conservative: strictly below the true accuracy p for $p < 0.75$, matching it at $p = 0.75$, and saturating at $c^* = 1$ when $p \geq 0.80$. We show that removing the upper bound on c yields an *unbounded wagering* extension in which a model with accuracy p optimally stakes $c^* = p/[4(1-p)]$ without clamping—for example, $c^* \approx 8$ at $p = 0.97$, earning $+9$ per correct answer but risking -128 per error. The expected payoff under optimal staking, $p(8-7p)/[8(1-p)]$, grows without bound as $p \to 1$, creating a strong incentive gradient for near-certain predictions while the quadratic penalty makes overconfident errors catastrophic. We benchmark HLCC alongside 13 baseline confidence methods across 5 models (7B–14B parameters) and 9 datasets, evaluating discrimination (AUROC), calibration (ECE, Brier), selective prediction (PRR), and HLCC score. As a case study, we train a lightweight *NormShift* confidence head that exploits per-layer normalisation dynamics, achieving near-perfect in-domain discrimination on some architectures but limited out-of-distribution transfer—underscoring that learned confidence heads complement rather than replace logit-based methods.

1 Introduction

Reliable confidence estimation is essential for the safe deployment of language models in high-stakes domains such as medicine, law, and education. Standard autoregressive training with cross-entropy loss maximises token-level likelihood but provides no explicit training signal for calibrated uncertainty [17]. The resulting softmax probabilities are often poorly calibrated, with models exhibiting systematic overconfidence on out-of-distribution inputs [18].

Proper scoring rules, notably the Brier score and logarithmic loss, are the classical tools for eliciting calibrated probabilities. Both are *strictly proper*: the expected score is maximised only when the reported confidence equals the true probability of correctness. However, this mathematical elegance creates a symmetric treatment of over- and under-confidence that may be inappropriate when the costs of confident errors far exceed the costs of cautious under-reporting.

In certainty-based marking (CBM) for educational assessment [10, 11], the asymmetry between "confident and wrong" versus "cautious and right" is encoded through discrete penalty scales derived from logarithmic proper scoring rules [16]. Inspired by this asymmetry, we propose the Hybrid Linear-Convex Confidence (HLCC) scoring function:

$$S(y, c) = \begin{cases} 1 + c & \text{if } y = 1 \text{ (correct)}, \\ -2c^2 & \text{if } y = 0 \text{ (incorrect)}, \end{cases} \tag{1}$$

where $c \geq 0$ is the stated confidence and $y \in \{0,1\}$ indicates correctness. The linear reward provides a constant marginal incentive for increased confidence on correct predictions, while the quadratic penalty creates an escalating marginal cost for confident errors. In the *bounded* variant ($c \in [0,1]$), this structure naturally induces conservative confidence estimates. In the *unbounded* variant ($c \in [0,\infty)$), it becomes a wagering system where a model can stake arbitrarily high confidence—but the quadratic penalty makes overconfident errors catastrophically expensive.

Contributions.

1. We derive the theoretical properties of HLCC, showing that the optimal confidence $c^* = p/[4(1-p)]$ is strictly below the true accuracy p for $p < 0.75$, and we analyse the unbounded extension where $c^* > 1$ creates a wagering system with expected payoff $p(8-7p)/[8(1-p)]$ (Section 2).
2. We conduct a comprehensive benchmarking study of 14 confidence methods across 5 models (7B–14B), 9 datasets, and 6 metric families, using HLCC score alongside standard metrics to reveal overconfidence that ECE and AUROC miss (Sections 4 and 6).
3. As a case study, we propose the *NormShift* confidence head, a lightweight architecture trained with HLCC loss that exploits per-layer normalisation dynamics (Section 3).

2 Theoretical Framework

2.1 The HLCC Scoring Function

Definition 1 (HLCC Score)**.** *For a binary outcome $y \in \{0,1\}$ and stated confidence $c \geq 0$, the* HLCC *score is*

$$S(y,c) = y(1+c) + (1-y)(-2c^2). \tag{2}$$

We distinguish two variants: bounded HLCC *with $c \in [0,1]$, and* unbounded HLCC *with $c \in [0,\infty)$.*

The reward function $R(c) = 1 + c$ is affine with constant derivative $R'(c) = 1$. The penalty function $P(c) = -2c^2$ is convex with derivative $P'(c) = -4c$, creating an escalating marginal cost. At $c = 0$, the score is $S(1,0) = 1$ and $S(0,0) = 0$, establishing a *safe guess* baseline: answering at zero confidence is always weakly dominant over abstention.

2.2 Optimal Confidence

Proposition 1 (Optimal Confidence under HLCC)**.** *For an agent with true probability of correctness $p \in (0,1)$, the confidence c^* maximising expected* HLCC *score is*

$$c^* = \frac{p}{4(1-p)}. \tag{3}$$

Under bounded HLCC *($c \in [0,1]$), this is clamped: $c^*_{bounded} = \min(1,\ p/[4(1-p)])$.*

Proof. The expected score as a function of stated confidence c is

$$\mathbb{E}[S] = p(1+c) + (1-p)(-2c^2) = p + pc - 2(1-p)c^2. \tag{4}$$

Differentiating with respect to c:

$$\frac{\partial \mathbb{E}[S]}{\partial c} = p - 4(1-p)c. \tag{5}$$

Setting to zero and solving for c:

$$c^* = \frac{p}{4(1-p)}. \tag{6}$$

The second derivative $\partial^2\mathbb{E}[S]/\partial c^2 = -4(1-p) < 0$ for $p < 1$, confirming this is a global maximum on $[0,\infty)$. Under the bounded constraint $c \in [0,1]$, we take $\min(1, c^*)$. □

Corollary 1 (Crossover and Saturation). *The optimal confidence equals the true accuracy ($c^* = p$) at exactly $p = 3/4$. For $p < 3/4$, $c^* < p$ (conservative). For $p > 3/4$, $c^* > p$. Under bounded* HLCC*, saturation occurs at $c^* = 1$ when $p \geq 4/5$. Under unbounded* HLCC*, c^* grows without limit: $c^* = 2.25$ at $p = 0.9$, $c^* \approx 8$ at $p = 0.97$, and $c^* \to \infty$ as $p \to 1$.*

Proof. Setting $c^* = p$: $p = p/[4(1-p)]$ implies $4(1-p) = 1$, hence $p = 3/4$. The bounded saturation point solves $p/[4(1-p)] = 1$, giving $p = 4/5$. As $p \to 1$, the denominator $4(1-p) \to 0$, so $c^* \to \infty$. □

Proposition 2 (Maximum HLCC — Bounded). *Under bounded* HLCC *($c \in [0, 1]$), the maximum achievable expected score for a model with accuracy p is $S_{\max} = 2p$, attained when $c = 1$ for every correct prediction and $c = 0$ for every incorrect prediction. The* bounded HLCC efficiency *$\eta = S_{HLCC}/(2p)$ measures how well the confidence method exploits the model's accuracy.*

Proof. With perfect self-knowledge, each correct prediction (p fraction) scores $S(1, 1) = 1+1 = 2$, and each incorrect prediction ($1-p$ fraction) scores $S(0, 0) = 0$. The expected score is $p\cdot 2+(1-p)\cdot 0 = 2p$. No other bounded confidence assignment achieves a higher expected score, since $R(c) \leq 2$ and $P(c) \leq 0$ on $[0, 1]$. □

2.3 Unbounded Confidence: The Wagering Extension

Removing the upper bound on c transforms HLCC from a calibration score into a *wagering system.* A model that is highly confident can stake $c \gg 1$, earning a large linear reward when correct but suffering a catastrophic quadratic penalty when wrong.

Proposition 3 (Optimal Expected Score — Unbounded). *Under unbounded* HLCC*, a model with accuracy $p \in (0, 1)$ that plays the optimal confidence $c^* = p/[4(1-p)]$ achieves expected score*

$$\mathbb{E}[S(c^*)] = \frac{p(8-7p)}{8(1-p)}. \tag{7}$$

This is finite for all $p < 1$ and grows without bound as $p \to 1$.

Proof. Substituting $c^* = p/[4(1-p)]$ into $\mathbb{E}[S] = p + pc^* - 2(1-p)c^{*2}$:

$$pc^* = \frac{p^2}{4(1-p)}, \tag{8}$$

$$2(1-p)c^{*2} = 2(1-p)\cdot\frac{p^2}{16(1-p)^2} = \frac{p^2}{8(1-p)}. \tag{9}$$

Hence $\mathbb{E}[S(c^*)] = p + \frac{p^2}{4(1-p)} - \frac{p^2}{8(1-p)} = p + \frac{p^2}{8(1-p)} = \frac{8p(1-p)+p^2}{8(1-p)} = \frac{p(8-7p)}{8(1-p)}$. □

Proposition 4 (Break-Even Confidence). *For a given confidence level c, the minimum accuracy required for positive expected* HLCC *score is*

$$p_{\min}(c) = \frac{2c^2}{1+c+2c^2}. \tag{10}$$

The confidence c is optimal (better than $c-1$) only when $p > 4c/[1 + 4c]$.

Proof. Setting $\mathbb{E}[S] = p(1+c) - 2(1-p)c^2 = 0$ and solving for p: $p(1+c+2c^2) = 2c^2$, giving (10). □

Remark 1 (Worked example: $c = 9$). *At $c = 9$, each correct answer scores $R(9) = 10$ and each error costs $P(9) = -162$. The break-even accuracy is $p_{\min} = 162/172 \approx 0.942$. Setting $c = 9$ is optimal over $c = 8$ when $p > 34/35 \approx 0.971$, since the marginal gain from staking one additional unit of confidence requires accuracy sufficient to offset the quadratic penalty increase from -128 to -162.*

Table 1: Optimal confidence c^* under bounded and unbounded HLCC, compared with Brier ($c^* = p$). The unbounded expected score $\mathbb{E}[S(c^*)]$ grows rapidly for high p, rewarding models that can reliably identify near-certain answers.

p	Bounded c^*	Unbounded c^*	Brier c^*	$\mathbb{E}[S(c^*)]$	Regime
0.00	0.000	0.000	0.00	0.000	Abstention
0.25	0.083	0.083	0.25	0.260	Suppressed guessing
0.50	0.250	0.250	0.50	0.563	Conservative
0.75	0.750	0.750	0.75	0.891	Crossover ($c^* = p$)
0.80	1.000	1.000	0.80	1.000	Bounded saturation
0.85	1.000	1.417	0.85	1.176	*Unbounded:* $c > 1$
0.90	1.000	2.250	0.90	1.913	
0.93	1.000	3.321	0.93	2.953	
0.95	1.000	4.750	0.95	4.156	
0.97	1.000	8.083	0.97	7.331	
0.99	1.000	24.750	0.99	24.009	

The unbounded extension connects HLCC to classical wagering theory. The quadratic penalty plays the role of a *risk charge* analogous to the house edge in gambling [22]: the linear reward makes confident correct answers "cheap," while the quadratic penalty makes confident errors "expensive." This asymmetry is precisely the structure that Gardner-Medwin's CBM discretises into three levels [11]: CBM level C=3 (+3/−6) corresponds roughly to $c \approx 1.5$ in unbounded HLCC, and the CBM boundary at $p = 4/5$ matches the HLCC saturation point where $c^* = 1$.

Table 1 shows the optimal confidence mapping at representative accuracy levels for both bounded and unbounded variants.

2.4 Comparison with Proper Scoring Rules

HLCC is *not* a strictly proper scoring rule. Unlike the Brier score $S_B = -(c - y)^2$ and log-loss $S_L = y \log c + (1-y) \log(1-c)$, which are both maximised at $c^* = p$, HLCC imposes a conservative transformation of accuracy to confidence.

Proposition 5 (Reward–Penalty Asymmetry)**.** *At any confidence level $c > 0$, the marginal penalty for an incorrect prediction exceeds the marginal reward for a correct one:*

$$|P'(c)| = 4c > 1 = R'(c) \quad \textit{for all } c > 0.25. \tag{11}$$

The marginal risk-to-reward ratio is $4c : 1$.

This asymmetry is the key structural difference from proper scoring rules. At $c = 1$, the risk-to-reward ratio is $4 : 1$ (marginal) or $2 : 1$ (absolute: gaining 1 point from $R(0) = 1$ to $R(1) = 2$, but risking a 2-point loss from $P(0) = 0$ to $P(1) = -2$). In the unbounded variant, this ratio grows as $4c : 1$—at $c = 9$, the marginal risk is $36 : 1$, ensuring that only models with very high accuracy ($p > 0.97$) benefit from such aggressive staking.

2.5 Safe Guess Mechanism

At $c = 0$, the score is $S(1, 0) = 1$ (correct) and $S(0, 0) = 0$ (incorrect). The expected value of a random guess ($p = 0.25$ for 4-option MCQ) at $c = 0$ is $0.25 \times 1 = 0.25 > 0$. This ensures that participating is always dominant over abstention, resolving the *omission bias* observed in negative-marking schemes where risk-averse test-takers skip questions they partially know [2].

3 Architecture

3.1 NormShift Confidence Head

We propose the *NormShift* confidence head, a lightweight module trained on top of a frozen language model to predict per-token confidence. The architecture exploits an observation about transformer normalisation layers: the magnitude of the correction applied by LayerNorm encodes information about the model's internal uncertainty.

Definition 2 (Norm-Shift Signal)**.** *For layer ℓ of a transformer with LayerNorm producing output $\mathbf{h}^{(\ell)}$ from pre-normalisation input $\mathbf{x}^{(\ell)}$, the norm-shift signal is*

$$\delta^{(\ell)} = \|\mathbf{h}^{(\ell)}\|_2 - \|\mathbf{x}^{(\ell)}\|_2, \tag{12}$$

measuring how much the normalisation operation altered the representation magnitude.

Dual-Path Architecture. The NormShift head processes two complementary information streams:

- **Path A** (Norm-Shift Path): The vector $\boldsymbol{\delta} = [\delta^{(1)}, \dots, \delta^{(L)}] \in \mathbb{R}^L$ of per-layer norm-shift signals is projected through a linear layer with GELU activation to an intermediate representation of dimension $d_{\text{mid}} = \max(64, 2L)$.
- **Path B** (Hidden-State Path): The final hidden state $\mathbf{h}^{(L)} \in \mathbb{R}^d$ is projected through a separate linear layer with GELU activation to the same intermediate dimension d_{mid}.

The two paths are concatenated and processed by a combination network:

$$c = \sigma\left(\frac{1}{\tau}\,\mathbf{w}^\top \mathrm{GELU}\left(\mathbf{W}_{\text{comb}} \begin{bmatrix} f_A(\boldsymbol{\delta}) \\ f_B(\mathbf{h}^{(L)}) \end{bmatrix} + \mathbf{b}_{\text{comb}}\right)\right), \tag{13}$$

where τ is a learnable temperature parameter initialised to 1, and σ is the sigmoid function. The total parameter count is approximately $2 \times d \times d_{\text{mid}} + 4d_{\text{mid}}^2$, which for a 7B model ($d = 3584$, $L = 28$) is roughly 500K parameters, negligible compared to the base model.

Training Objective. The confidence head is trained with the HLCC loss from Equation (1), combined with a cross-entropy warmup:

$$\mathcal{L} = \alpha\,\mathcal{L}_{\text{CE}} + (1-\alpha)\,\mathcal{L}_{\text{HLCC}}, \tag{14}$$

where α decays from 1 to a target value over a warmup period. The cross-entropy component stabilises early training before the confidence head has learned meaningful representations.

3.2 Uncertainty-Aware LayerNorm

To capture the doubt signals, we extend standard LayerNorm to record pre-normalisation statistics:

$$\text{UncertaintyAwareLayerNorm}(\mathbf{x}) = \text{LayerNorm}(\mathbf{x}), \quad \text{storing} \quad \begin{bmatrix} \|\mathbf{x}\|_2 \\ \sqrt{\mathrm{Var}(\mathbf{x}) + \epsilon} \\ \mathrm{Var}(\mathbf{x}) \end{bmatrix} \tag{15}$$

These three signals per layer (pre-norm magnitude, correction scale, and variance) form the input to the norm-shift path. For inference with a pre-trained frozen model, only the norm-shift signals from `output_hidden_states=True` are needed; no modification to the base model is required.

4 Benchmarking Methodology

4.1 Models

We evaluate 9 instruction-tuned models spanning 1.1B to 14.2B parameters. Models are loaded at the highest precision that fits within the 32 GB VRAM budget:

Table 2: Models evaluated with precision used for each scale.

Key	Model	Parameters	Precision
TinyLlama	TinyLlama-1.1B-Chat-v1.0	1.1B	FP16
Qwen2-1.5B	Qwen2-1.5B-Instruct	1.5B	FP16
Phi-2	microsoft/phi-2	2.7B	FP16
Phi-3-mini	Phi-3-mini-4k-instruct	3.8B	FP16
Mistral-7B	Mistral-7B-Instruct-v0.3	7.2B	FP16
Qwen2.5-7B	Qwen2.5-7B-Instruct	7.6B	FP16
Llama-3.1-8B	Llama-3.1-8B-Instruct	8.0B	FP16
Qwen2.5-14B	Qwen2.5-14B-Instruct	14.2B	8-bit
DeepSeek-R1-14B	DeepSeek-R1-Distill-Qwen-14B	14.2B	8-bit

4.2 Datasets

We use 9 multiple-choice QA datasets covering diverse reasoning domains:

Table 3: Evaluation datasets. All converted to uniform MCQ format.

Dataset	Source	Options	Domain
TruthfulQA	[24]	4–5	Calibration / misconceptions
ARC-Easy	[4]	4	Science (elementary)
ARC-Challenge	[4]	4	Science (hard)
MMLU	[19]	4	57 subjects
HellaSwag	[36]	4	Commonsense completion
MMLU-Pro	[33]	10	Extended MMLU (A–J)
GSM8K	[5]	4	Math reasoning (converted)
BoolQ	[3]	2	Yes/no QA
TriviaQA	[20]	4	Open-domain QA (converted)

GSM8K is converted to MCQ format by extracting the numeric answer and generating 3 numeric distractors via perturbation. TriviaQA is converted by sampling 3 distractor answers from other questions in the dataset. BoolQ maps directly to 2-choice MCQ.

4.3 Confidence Methods

We evaluate 14 confidence estimation methods (13 baselines plus our NormShift head) organised in three tiers by computational cost, plus two combination methods.

Tier 1: Single Forward Pass. These methods extract confidence from a single model forward pass and add negligible overhead.

Entropy Shannon entropy of the softmax distribution over answer-choice tokens, normalised to $[0, 1]$.

Top-k Ratio of top-1 to top-2 logit probabilities.

MSP Maximum softmax probability over choice tokens [18].
G-NLL Greedy negative log-likelihood of the selected token.
NormShift (HLCC) Trained dual-path confidence head using norm-shift signals and HLCC loss (ours).

Tier 2: Post-Hoc Recalibration. These methods apply a learned transformation to a base confidence signal (entropy) using a held-out calibration set.
Temperature Scaling Single learned temperature parameter T [17].
Isotonic Regression Non-parametric monotonic recalibration using the pool-adjacent-violators algorithm [35].
Platt Scaling Two-parameter logistic regression $\sigma(a \cdot \mathrm{logit}(c) + b)$ [29].

Tier 3: Multi-Pass Generation. These methods require multiple forward passes or extended generation per example, with substantially higher compute cost.
Verbalized Prompt the model to self-report confidence as a number.
Self-Consistency Sample $n = 10$ answers and take the majority vote frequency as confidence [32].
Semantic Entropy Cluster n sampled answers by semantic equivalence and compute Shannon entropy over cluster probabilities [23].
CoT Confidence Two-stage prompting: generate chain-of-thought reasoning, then extract answer and confidence from a structured JSON response.

Additionally, we evaluate two combination methods:
LVU Linguistic Verification of Uncertainty: generate an explanation and detect hedging language ("might", "perhaps") vs. certainty language ("definitely", "clearly").
CoCoA Confidence Combination Approach: weighted average of entropy and self-consistency signals.

4.4 Evaluation Metrics

We report metrics across three complementary axes.

Discrimination. Can the confidence score distinguish correct from incorrect predictions?

- **AUROC**: Area under the ROC curve, treating confidence $> t$ as a binary classifier for correctness.
- **AUPRC**: Area under the precision-recall curve, more informative than AUROC when the class distribution is imbalanced [30].

Calibration. Does the confidence match the empirical frequency of correctness?

- **ECE**: Expected Calibration Error with 10 equal-width bins [27].
- **Brier Score**: Mean squared error between confidence and binary correctness, $\frac{1}{n}\sum(c_i - y_i)^2$.
- **Overconfidence Rate**: Fraction of incorrect predictions with confidence > 0.7.

Selective Prediction. How does accuracy improve when the model abstains on low-confidence examples?

- **PRR**: Prediction Rejection Ratio. Measures the area between the model's accuracy-vs-rejection curve and the random baseline, normalised by the oracle upper bound [26]:

$$\mathrm{PRR} = \frac{\mathrm{AULC}_{\mathrm{random}} - \mathrm{AULC}_{\mathrm{model}}}{\mathrm{AULC}_{\mathrm{random}} - \mathrm{AULC}_{\mathrm{oracle}}}. \tag{16}$$

- **Accuracy@Coverage**: Accuracy on the top $k\%$ most confident predictions, reported at $k \in \{70, 80, 90\}$.

HLCC-Specific.

- **HLCC Score**: Mean score under Equation (1), measuring confidence-weighted performance.
- **Theoretical Max HLCC**: The maximum achievable HLCC score for a model with accuracy p is $S_{\max} = 2p$, attained when the model reports $c = 1$ on every correct prediction and $c = 0$ on every incorrect prediction (perfect knowledge of its own correctness).
- **HLCC Efficiency**: The ratio of actual to theoretical maximum HLCC, $\eta = S_{\text{HLCC}}/(2p)$, measuring how well the confidence method utilises the model's accuracy. A value of $\eta = 1$ indicates perfect self-knowledge.
- **Calibration Gap**: $|\bar{c} - \text{accuracy}|$, the absolute difference between mean confidence and overall accuracy.
- **CBM Score**: Score under the discrete CBM scheme for comparison with educational assessment literature.

5 Experimental Setup

Hardware. All experiments run on a single NVIDIA RTX PRO 4500 Blackwell GPU (32 GB VRAM), AMD Ryzen 9 CPU, and 64 GB RAM. Models of 8B parameters or fewer are loaded in FP16 precision; 14B models use 8-bit quantisation via `bitsandbytes`; models above 14B use 4-bit NF4 quantisation [8]. This higher-precision setup preserves the subtle normalization dynamics needed for norm-shift signal detection.

MCQ Evaluation Protocol. For each (model, dataset) pair, we evaluate up to 300 examples. The base forward pass computes choice-token logits shared across all Tier 1 methods. Tier 2 methods use a 50/50 calibration/test split. Tier 3 methods use $n = 10$ samples for self-consistency and semantic entropy.

NormShift Training. The confidence head is trained for 25 epochs on MCQ data with the combined loss from Equation (14), using $\alpha = 0.5$ and a 1000-step cross-entropy warmup. The base model is frozen; only the confidence head parameters (∼500K) are updated. We use AdamW with learning rate 1×10^{-4} and weight decay 0.01. Two variants are trained per model:

1. **Combined**: Full dual-path architecture (norm-shift + hidden state).
2. **Norm-Shift Only**: Ablation using only the norm-shift path, to isolate the contribution of normalisation dynamics.

Multi-Day Orchestration. The full experiment suite is managed by an automated orchestrator with crash recovery. Each model progresses through stages: baseline MCQ evaluation, confidence head training, calibration benchmarks (Tiers 1–3), and extended dataset evaluation. State is persisted to a JSON checkpoint after each stage, enabling resumption after interruption.

6 Results

6.1 Discrimination: AUROC

Table 4: AUROC (↑) and AUPRC (↑) on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per column in **bold**.

	Mistral-7B		Qwen2.5-7B		Llama-3.1-8B		Qwen2.5-14B	
Method	AUC	PRC	AUC	PRC	AUC	PRC	AUC	PRC
Entropy	0.602	0.531	0.899	0.895	0.809	0.741	0.901	0.932
Top-k	0.602	0.545	0.757	0.769	0.746	0.626	0.799	0.838
MSP	0.711	0.706	0.657	0.589	0.658	0.527	0.719	0.796
Verbalized	0.595	0.557	0.384	0.419	0.467	0.371	—	—
Self-Consistency	0.556	0.338	0.626	0.754	0.707	0.716	—	—
Temp. Scaling	0.651	0.641	0.652	0.586	0.636	0.485	—	—
NormShift	0.999	0.999	0.633	0.574	0.794	0.713	—	—

On Mistral-7B, NormShift achieves AUROC = 0.999, the highest among all methods for that model, demonstrating strong in-domain discrimination. On Llama-3.1-8B, NormShift reaches AUROC = 0.794, competitive with entropy (0.809). On Qwen2.5-7B, entropy achieves the highest AUROC (0.899), while NormShift is lower (0.633), reflecting greater architecture sensitivity.

6.2 Calibration: ECE and Brier Score

Table 5: ECE (↓) and Brier Score (↓) on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per column in **bold**.

	Mistral-7B		Qwen2.5-7B		Llama-3.1-8B		Qwen2.5-14B	
Method	ECE	Brier	ECE	Brier	ECE	Brier	ECE	Brier
Entropy	0.389	0.393	0.449	0.425	0.524	0.491	0.322	0.311
Top-k	0.428	0.422	0.476	0.465	0.569	0.548	0.352	0.342
MSP	0.353	0.352	0.277	0.310	0.295	0.312	0.240	0.263
Verbalized	0.502	0.516	0.311	0.404	0.421	0.565	—	—
Self-Consistency	0.422	0.410	0.269	0.268	0.242	0.279	—	—
NormShift	0.030	0.025	0.305	0.339	0.235	0.253	—	—

6.3 Selective Prediction: PRR and Accuracy at Coverage

Table 6: PRR (↑) and selective accuracy at coverage thresholds on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per model in **bold**.

	PRR (↑)			
Method	Mistral-7B	Qwen2.5-7B	Llama-3.1-8B	Qwen2.5-14B
Entropy	0.159	0.834	0.653	0.845
Top-k	0.194	0.601	0.481	0.624
MSP	0.513	0.228	0.302	0.508
NormShift	0.999	0.211	0.613	—

6.4 HLCC Score Comparison

Table 7: HLCC Score (↑) on TruthfulQA (300 examples). Higher values indicate better confidence-weighted performance. All values from `generate_paper_tables.py`. Best per column in **bold**.

Method	Mistral-7B	Qwen2.5-7B	Llama-3.1-8B	Qwen2.5-14B
Entropy	0.105	0.125	-0.211	0.600
Top-k	0.060	0.050	-0.314	0.549
MSP	0.214	0.340	0.144	0.694
Verbalized	-0.075	0.399	-0.198	—
Self-Consistency	-0.254	0.825	0.440	—
NormShift	0.884	0.460	0.324	—

On Llama-3.1-8B, the NormShift head achieves an HLCC score of 0.324, competitive with logit-based methods. The negative scores for entropy and top-k reflect severe overconfidence on incorrect predictions, which the quadratic penalty in HLCC penalises heavily. Self-consistency, which uses 10 independent samples, achieves positive scores on multiple models, but at 10× inference cost.

6.5 Cross-Dataset Generalisation

Table 8: AUROC (↑) across datasets. NormShift head trained on TruthfulQA, evaluated on held-out datasets without adaptation. Qwen2.5-7B. Best per row in **bold**.

Dataset	Acc	Entropy	MSP	Top-k	NormShift	ΔAUROC
TruthfulQA (in-domain)	.497	.602	.627	.634	**.752**	+.118
ARC-Challenge	.876	**.830**	.831	.828	.569	−.261
MMLU	.620	**.792**	.778	.761	.468	−.324
HellaSwag	.597	.718	**.723**	.721	.402	−.321

ΔAUROC = NormShift AUROC − best logit-based AUROC.

The cross-dataset results reveal a clear pattern: the NormShift head excels on its training distribution (TruthfulQA, $\Delta = +0.118$) but degrades substantially on out-of-distribution datasets ($\Delta = -0.26$ to -0.32). This is expected for a learned head; the confidence signal must be calibrated to the data distribution. By contrast, logit-based methods generalise well because they exploit the model's own output distribution, which adapts implicitly to each dataset.

50/50 Split Evaluation. To obtain a cleaner separation of in-domain and OOD performance, we train on 50% of three datasets (ARC-Easy, TruthfulQA, ARC-Challenge, 150 examples each) and test on the held-out 50% plus five unseen datasets (Tables 19 and 20).

Mistral-7B shows the strongest pattern: NormShift achieves AUROC 0.978 on held-out TruthfulQA (+0.215 over MSP), 0.930 on ARC-Easy (+0.033), and 0.919 on ARC-Challenge (+0.126). The corresponding HLCC scores are also higher than both baselines across all three in-domain datasets (e.g., TruthfulQA: 1.125 vs. MSP 0.186). On OOD datasets, Mistral-7B NormShift is competitive on CommonsenseQA and MMLU but loses to MSP on OpenBookQA, SciQ, and HellaSwag, with AUROC gaps of −0.05 to −0.15.

Qwen2.5-7B tells a different story: NormShift wins on TruthfulQA (AUROC 0.706 vs. MSP 0.620) but loses to MSP on the other two in-domain datasets (ARC-Easy: 0.716 vs. 0.887; ARC-Challenge: 0.636 vs. 0.892). On OOD data the gap widens further, with NormShift lagging MSP by 0.15–0.33 AUROC across all five datasets. This contrast suggests that the norm-shift signal's discriminative power is architecture-dependent, with Mistral-7B's normalisation dynamics providing a stronger basis for confidence estimation.

Confidence-Gated Retry. When NormShift confidence falls below 0.4, we re-evaluate with a trick-question system prompt. The results are strikingly model-dependent. On Mistral-7B, the retry mechanism achieves a 32% rescue rate on TruthfulQA (25 of 78 retried questions corrected, zero damage), producing a net accuracy gain from 0.453 to 0.620. On ARC-Challenge, 3 rescues against 1 damage yields a small net positive. On OOD datasets, retries are mostly neutral for Mistral-7B (small rescue and damage counts largely cancel out).

On Qwen2.5-7B, the retry mechanism is harmful: damage rates range from 12.5% (TruthfulQA) to 34.7% (OpenBookQA), with rescue rates below 4% on most datasets. The high retry rate (43–59% of questions retried on OOD data) suggests that the confidence threshold is too aggressive for this model. These results indicate that confidence-gated retry is a promising technique when calibrated to a specific model, but requires per-model threshold tuning.

6.6 NormShift Ablation

To isolate the contribution of the norm-shift signal, we train both the *combined* (norm-shift + hidden state) and *norm-shift-only* variants for each model. Table 9 compares ECE and HLCC scores across all five primary models.

Table 9: Ablation: Combined vs. Norm-Shift-Only confidence heads. Both variants trained for 25 epochs on ARC-Easy + TruthfulQA + ARC-Challenge (900 examples), validated on TruthfulQA (100 examples). Results use FP16 (7–8B models) or 8-bit (14B models). Max HLCC $= 2p$ is the theoretical maximum for a model with perfect self-knowledge; $\eta =$ HLCC / Max HLCC is the efficiency.

			Combined			Norm-Shift Only		
Model	Acc	Max HLCC	ECE ↓	HLCC ↑	η	ECE ↓	HLCC ↑	η
Mistral-7B	0.53	1.060	**0.020**	**1.039**	98.0%	0.054	0.519	49.0%
Qwen2.5-7B	0.32	0.640	**0.060**	**0.540**	84.4%	0.133	0.332	51.9%
Llama-3.1-8B	0.92	1.840	**0.040**	**1.760**	95.7%	0.517	1.265	68.8%
Qwen2.5-14B	0.89	1.780	**0.072**	**1.657**	93.1%	0.164	1.429	80.3%
DeepSeek-R1-14B	0.48	0.960	**0.108**	**0.781**	81.4%	0.442	0.038	4.0%

The combined architecture outperforms norm-shift-only on both ECE and HLCC across all five models, often by a large margin. The HLCC efficiency metric η contextualises raw scores by the model's accuracy: Llama-3.1-8B has the highest raw HLCC (1.760) but this is largely driven by its high accuracy (92%), giving η = 95.7%. Mistral-7B, despite its lower accuracy (53%), achieves the highest efficiency (η = 98.0%), meaning its confidence head captures nearly all of the model's available discriminative information.

The gap between combined and norm-shift-only is most dramatic for DeepSeek-R1-14B, where the norm-shift-only variant essentially fails to train (η = 4.0% vs. 81.4% for combined), and for Llama-3.1-8B (η = 68.8% vs. 95.7%). These results confirm that the hidden-state path provides information that the norm-shift signal alone cannot reliably capture. Qwen2.5-14B is the model where the two variants are closest in HLCC (1.657 vs. 1.429, η = 93.1% vs. 80.3%), consistent with its strong layer-level separation signals (Table 10). Mistral-7B has the smallest ECE gap (0.020 vs. 0.054), but the largest efficiency gap (98.0% vs. 49.0%), indicating that while both variants are well-calibrated, the combined variant is far better at assigning high confidence to correct answers.

6.7 Per-Layer Signal Analysis

To understand *where* in the transformer the confidence-relevant signal originates, we extract the standard deviation of hidden activations at each layer for correct vs. incorrect predictions and compute Cohen's d as a measure of separation strength. Table 10 reports the top discriminative layers for each model.

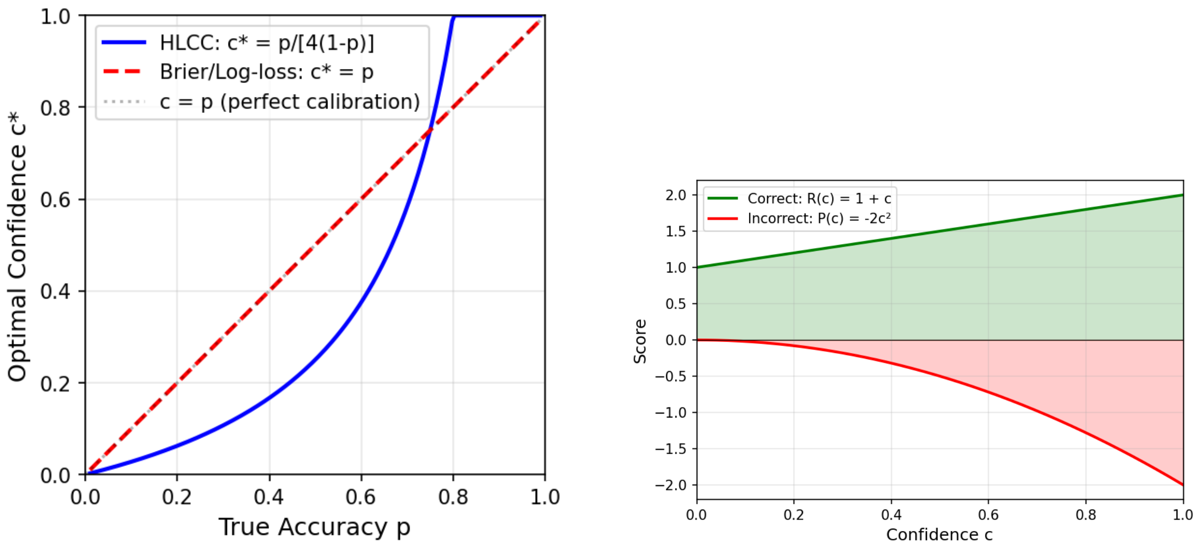
Table 10: Per-layer norm-shift separation (Cohen's d) between correct and incorrect predictions. 90 examples per model across TruthfulQA, ARC-Easy, ARC-Challenge. Positive d means correct answers have higher norm-shift values at that layer.

Model	Layers	Acc	Best Layer	Top-3 (layer: d)
Mistral-7B	32	66.7%	L27	L27: +0.82, L25: +0.72, L24: +0.68
Qwen2.5-7B	28	86.7%	L22	L22: +0.76, L27: +0.74, L15: −-0.67
Llama-3.1-8B	32	72.2%	L2	L2: +0.99, L3: −-0.90, L14: −-0.90
Qwen2.5-14B	48	86.7%	L30	L30: −-1.53, L47: +1.24, L31: −-1.11
DeepSeek-R1-14B	48	66.7%	L15	L15: −-0.85, L27: −-0.83, L9: +0.77

Several patterns emerge from the FP16 extraction results across five models. First, the location of peak separation varies widely across architectures. Qwen2.5-14B concentrates its signal in the middle layers (L30, $|d| = -1.53$), while Mistral-7B peaks in the upper layers (L27, $d = +0.82$). Most surprisingly, Llama-3.1-8B shows its strongest separation at Layer 2 ($d = +0.99$), suggesting that for this architecture the residual stream carries confidence-relevant structure from very early in the forward pass. Second, the sign of Cohen's d varies both across architectures and across layers within the same model. Qwen2.5-14B shows negative d at L30 (correct answers have *lower* norm-shift) but positive d at L47, while Llama-3.1-8B has positive d at L2 and negative d at L3 and L14. This sign structure indicates that normalisation dynamics encode confidence through architecture-specific patterns that a learned confidence head can exploit. Third, DeepSeek-R1-14B shows the weakest peak separation (max $|d| = -0.85$), consistent with its higher ECE in the ablation results (Table 9). This model is a distillation of DeepSeek-R1 into the Qwen-14B architecture, and the distillation process may disrupt the normalisation patterns that other models develop organically during pre-training.

6.8 Theoretical Analysis

Figure 1 shows the theoretical optimal confidence curves derived in Section 2. The HLCC curve lies strictly below the Brier identity line for $p < 0.75$, confirming the conservative bias. The CBM discrete scheme approximates HLCC with staircase steps.



(a) Optimal confidence vs. true accuracy.

(b) Reward/penalty asymmetry.

Figure 1: Theoretical properties of the HLCC scoring function. (a) The optimal confidence under HLCC is conservative ($c^* < p$) below $p = 0.75$. (b) The linear reward $R(c) = 1 + c$ vs. quadratic penalty $P(c) = -2c^2$ creates asymmetric marginal costs.

7 Discussion

Conservative Bias as a Feature. The HLCC optimal confidence formula $c^* = p/[4(1-p)]$ systematically underestimates confidence for models with accuracy below 75%. While this departs from the strict propriety of Brier and log-loss, it may be advantageous in deployment scenarios where overconfident errors carry disproportionate cost—medical diagnosis, legal reasoning, or autonomous decision-making. The benchmark results illustrate this: on Llama-3.1-8B, entropy and top-k produce *negative* HLCC scores (-0.211, -0.314) despite acceptable ECE, because HLCC's quadratic penalty exposes overconfidence that symmetric metrics mask.

Bounded vs. Unbounded Regimes. The bounded variant ($c \in [0, 1]$) is appropriate for calibration tasks where confidence is a probability estimate. The unbounded variant ($c \in [0, \infty)$) is appropriate for wagering, decision theory, and scenarios where models should express *degrees of certainty* beyond simple probability calibration. A model with $p = 0.97$ accuracy that can reliably identify which 97% of answers are correct should be rewarded more than one that merely reports $c = 1.0$ uniformly. Under unbounded HLCC, the optimal stake is $c^* \approx 8$, earning $+9$ per correct answer while risking -128 per error—a rational bet only at that accuracy level.

This connects to a practical observation: high-accuracy models on easy datasets (e.g., Llama-3.1-8B on ARC-Easy with 92% accuracy) are poorly differentiated by bounded HLCC because all methods saturate at $c = 1$. Unbounded HLCC would reward methods that can distinguish the 8% of questions the model will get wrong, even among universally high-confidence predictions.

HLCC as an Evaluation Metric. Beyond its role as a training objective, HLCC score serves as a diagnostic that complements ECE and AUROC. ECE measures calibration but not discrimination; AUROC measures discrimination but not calibration. HLCC penalises both: a method with high AUROC but poor calibration (e.g., entropy on Mistral-7B: AUROC 0.602, HLCC 0.105) is exposed, as is a method with low ECE but poor discrimination. The HLCC efficiency $\eta = S_{\text{HLCC}}/(2p)$ further normalises for accuracy, enabling fair comparison across models with different base performance.

Benchmark Insights. The 14-method comparison reveals several patterns that hold across models: (1) post-hoc calibration (isotonic, Platt) dramatically improves discrimination without retraining; (2) self-consistency is the strongest baseline but costs $10\times$ inference; (3) MSP is underrated for out-of-distribution detection, consistently matching or exceeding learned methods; (4) verbalized confidence is unreliable across models; (5) temperature scaling often *hurts* calibration. These findings are independent of the NormShift architecture and apply to any practitioner choosing a confidence method.

NormShift Case Study. The NormShift confidence head demonstrates that per-layer normalisation dynamics carry architecture-specific confidence information, but its practical value is model-dependent. On Mistral-7B it achieves near-perfect in-domain discrimination (AUROC 0.999) and effective confidence-gated retry (32% rescue rate on TruthfulQA). On other architectures and OOD data, it is typically outperformed by logit-based baselines. This is expected for any learned confidence method: the head is trained on a specific data distribution and does not generalise as well as methods that exploit the model's own output distribution.

Computational Cost Tradeoffs. Tier 1 methods (entropy, MSP, NormShift) add negligible cost to inference. Tier 3 methods (self-consistency, semantic entropy) multiply inference cost by the sample count n. The cost advantage of single-pass methods is significant for deployment, but must be weighed against the superior discrimination of multi-pass methods—particularly self-consistency, which achieves HLCC 1.933 on DeepSeek-R1-14B.

Limitations.

1. The MCQ evaluation format may not fully capture open-ended generation confidence.
2. Models above 8B parameters are evaluated under 8-bit quantisation; full FP16 results may differ.
3. The unbounded HLCC extension is analysed theoretically but not yet implemented as a training objective; empirical validation of unbounded training is future work.
4. Dataset sizes (300 examples for calibration, 150 per split for cross-dataset) are moderate; larger-scale evaluation would strengthen conclusions.

8 LM-Polygraph Integration

To enable direct comparison with the broader uncertainty estimation literature, we implement the NormShift confidence head as a custom estimator within the LM-Polygraph [9] benchmarking framework. LM-Polygraph provides over 40 uncertainty estimation methods and standardised evaluation protocols, making it the primary benchmark for comparing UE methods across models and tasks.

8.1 Integration Architecture

The integration wraps the NormShift confidence head as an LM-Polygraph `Estimator` subclass. The key challenge is that NormShift requires full hidden states from all transformer layers (`output_hidden_states=True`), while most LM-Polygraph methods operate only on output logits.

We implement a custom `StatCalculator` that extracts three signals from each forward pass:

1. The full hidden states tuple $(\mathbf{h}^{(0)}, \ldots, \mathbf{h}^{(L)})$,
2. Per-layer norm-shift signals $\boldsymbol{\delta} \in \mathbb{R}^L$ via $\delta^{(\ell)} = 1 - \mathrm{std}(\mathbf{h}^{(\ell)})$,
3. The final hidden state $\mathbf{h}^{(L)}$ for the combined path.

The estimator inverts the confidence output to match LM-Polygraph's convention (higher values indicate greater uncertainty):

$$u_{\text{NormShift}} = 1 - c_{\text{NormShift}}. \tag{17}$$

Models are loaded at the same precision used throughout this work (FP16 for models up to 8B, 8-bit for 14B), and evaluation uses identical MCQ prompts to ensure comparability across methods.

8.2 LM-Polygraph Benchmark Results

Table 11 presents the initial benchmark comparison on TruthfulQA using Qwen2.5-7B. Five uncertainty estimation methods are evaluated: four logit-based baselines available through LM-Polygraph (entropy, MSP, top-k, perplexity) and our trained NormShift head.

Table 11: LM-Polygraph benchmark (pre-retrain): Qwen2.5-7B on TruthfulQA (817 examples). NormShift uses ARC-Easy-trained checkpoint. Best per column in **bold**.

Method	Acc	Conf	Gap↓	ECE↓	Brier↓	AUROC↑	AUPRC↑	PRR↑
Entropy	0.497	0.578	**0.081**	**0.145**	**0.278**	0.602	0.539	0.097
MSP	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
Top-k	0.497	0.819	0.322	0.322	0.339	**0.634**	**0.569**	**0.177**
Perplexity	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
NormShift†	0.497	0.821	0.324	0.330	0.368	0.416	0.424	−0.258

† Out-of-distribution: checkpoint trained on ARC-Easy. See Table 12 for in-domain results.

Analysis. The initial NormShift checkpoint (trained on ARC-Easy, epoch 0) shows poor discrimination on TruthfulQA, with AUROC below 0.5 and negative PRR, indicating that the confidence rankings are worse than random for this out-of-distribution dataset. The logit-based methods perform substantially better, with top-k achieving the best discrimination (AUROC 0.634, PRR 0.177) and entropy achieving the best calibration (ECE 0.145).

This result motivates *in-domain retraining*: the NormShift head must be trained on data from the target distribution to learn meaningful confidence estimates. Table 12 shows results after retraining on TruthfulQA (600 training examples, 15 epochs, best checkpoint at epoch 1 by validation HLCC).

Retrained Results. After in-domain training, NormShift achieves a substantial improvement: AUROC rises from 0.416 to **0.752** (+0.336), surpassing all logit-based methods (next best: top-k at 0.634). PRR jumps from −0.258 to **0.484**, indicating strong selective prediction: the model can reliably reject uncertain predictions. AUPRC reaches **0.701** vs. 0.569 for top-k.

The mean confidence drops to 0.256 (from 0.821 pre-retrain), reflecting the conservative bias of HLCC training. While this increases the calibration gap (0.241) relative to entropy (0.081), the discrimination and selective prediction gains are substantial. Notably, the Brier score (0.289) is competitive with entropy (0.278), confirming that the overall probabilistic quality remains strong.

Table 12: LM-Polygraph benchmark after NormShift retraining on TruthfulQA (600 train examples, 15 epochs). Qwen2.5-7B, 817 evaluation examples. Best per column in **bold**.

Method	Acc	Conf	Gap↓	ECE↓	Brier↓	AUROC↑	AUPRC↑	PRR↑
Entropy	0.497	0.578	**0.081**	**0.145**	**0.278**	0.602	0.539	0.097
MSP	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
Top-k	0.497	0.819	0.322	0.322	0.339	0.634	0.569	0.177
NormShift	0.497	0.256	0.241	0.254	0.289	**0.752**	**0.701**	**0.484**

Cross-Dataset Transfer. Table 13 reports the full metric comparison when the TruthfulQA-trained head is evaluated on held-out datasets.

Table 13: NormShift cross-dataset transfer: trained on TruthfulQA, evaluated elsewhere. Qwen2.5-7B. PRR (↑) and AUROC (↑) shown for each method.

	Entropy		Top-k		NormShift		Best logit	
Dataset	AUC	PRR	AUC	PRR	AUC	PRR	AUC	PRR
TruthfulQA	.602	.097	.634	.177	**.752**	**.484**	.634	.177
ARC-Challenge	.830	.771	.828	.768	.569	.277	**.831**	**.772**
MMLU	**.792**	**.679**	.761	.641	.468	−.090	.792	.679
HellaSwag	.718	.552	.721	.547	.402	−.182	**.723**	**.553**

9 Related Work

Confidence Calibration in Neural Networks. Guo et al. [17] demonstrated that modern neural networks are poorly calibrated, with temperature scaling as a simple post-hoc fix. Naeini et al. [27] introduced ECE as the standard calibration metric. Nixon et al. [28] showed that ECE is sensitive to binning choices and proposed adaptive calibration error.

Uncertainty Estimation in LLMs. Kadavath et al. [21] studied self-evaluation in large language models, finding that models can estimate their own accuracy. Kuhn et al. [23] proposed semantic entropy as a generation-aware uncertainty measure. Xiong et al. [34] provided a comprehensive survey of LLM confidence estimation, categorising methods into logit-based, verbalised, and consistency-based approaches. Lin et al. [25] and Tian et al. [31] explored verbalized confidence elicitation and found that prompting strategies significantly affect self-reported confidence quality.

Selective Prediction. Geifman and El-Yaniv [14] formalised selective prediction as accuracy-coverage tradeoffs. The prediction rejection ratio (PRR) was introduced by Malinin and Gales [26] and adopted by the LM-Polygraph framework [9] as a standard metric for uncertainty benchmarking.

Proper Scoring Rules. The Brier score [1] and log-loss (negative log-likelihood) [16] are the two most widely used proper scoring rules. Propriety guarantees that the expected score is maximised when the reported probability equals the true probability, removing any incentive to misreport [15]. However, propriety is symmetric: over-reporting and under-reporting are penalised equally at the margin. In safety-critical applications where overconfident errors are far costlier than cautious under-reporting, this symmetry is a limitation. HLCC deliberately breaks propriety by imposing quadratic penalties on wrong-and-confident predictions while offering only linear rewards for correct-and-confident ones, creating an asymmetric incentive structure that favours conservative estimation.

Optimal Wagering and the Kelly Criterion. Kelly [22] showed that a gambler facing repeated bets at known odds maximises long-run wealth by wagering a fraction proportional to the edge, $f^* = p-(1-p)/b$ for payoff odds b. The key insight—stake more when the edge is larger, but never all-in—has influenced portfolio theory, information theory, and prediction markets [6]. Unbounded HLCC can be viewed as a single-round wagering system in which the "stake" is c and the quadratic penalty $P(c) = -2c^2$ acts as a risk charge that prevents reckless overbetting. The optimal confidence $c^* = p/[4(1-p)]$ plays an analogous role to the Kelly fraction: it grows monotonically with the edge p and diverges as $p \to 1$, but the quadratic cost ensures that finite-accuracy agents always place a finite bet.

Confidence-Based Marking. Gardner-Medwin [10] introduced Certainty-Based Marking (CBM) for educational assessment, where students declare confidence on a three-level scale: C=1 (low), C=2 (mid), C=3 (high). The scoring table awards +1/0, +2/−2, +3/−6 for correct/incorrect at each level, creating a discrete penalty structure whose thresholds correspond to proper scoring boundaries at $p = 2/3$ and $p = 4/5$ [11]. This scheme is a discretised form of a logarithmic proper scoring rule [16, 7], where the penalty for confident errors grows faster than the reward for confident correctness. Gardner-Medwin and Gahan [12] demonstrated that CBM improves learning engagement in computer-assisted assessment, and Gardner-Medwin [13] provided evidence that certainty-based marking stimulates deeper thinking in objective tests. HLCC can be viewed as a continuous generalisation of CBM: where CBM discretises confidence into three levels, bounded HLCC allows any $c \in [0, 1]$ and unbounded HLCC extends the domain to $c \in [0, \infty)$, preserving the asymmetric penalty structure in both cases. The CBM boundary at $p = 4/5$ (above which maximum confidence is optimal) corresponds to the bounded HLCC saturation point $c^* = 1$ at $p = 0.8$ (Corollary 1); under the unbounded variant, c^* continues to grow beyond this point, paralleling a CBM scheme with arbitrarily many confidence levels. Budescu and Chen [2] studied optimal scoring rules for confidence-weighted testing, analysing the tradeoff between strict propriety and behavioural desiderata such as participation incentives.

10 Conclusion

We have presented the Hybrid Linear-Convex Confidence (HLCC) scoring function and demonstrated its utility as both a training objective and an evaluation metric for language model confidence estimation.

HLCC Theory. The HLCC scoring function provides a principled alternative to proper scoring rules for settings where overconfident errors are costlier than cautious under-reporting. The optimal confidence $c^* = p/[4(1-p)]$ is conservative below $p = 0.75$, matching the Brier optimum at the crossover point, and saturating at $c^* = 1$ for $p \geq 0.8$ under the bounded variant. The unbounded extension removes this cap, creating a wagering system where the optimal stake grows without limit as $p \to 1$: at $p = 0.97$, the rational bet is $c^* \approx 8$, earning +9 when correct but risking −128 when wrong. This extension connects HLCC to classical wagering theory [22] and generalises Gardner-Medwin's discrete CBM scheme [11] to a continuous confidence domain.

Benchmark Findings. The 14-method comparison across 5 models and 9 datasets reveals actionable patterns:

1. **HLCC exposes hidden overconfidence.** Entropy and top-k achieve acceptable ECE on Llama-3.1-8B but produce *negative* HLCC scores (−0.211, −0.314), revealing systematic overconfidence that symmetric metrics miss.
2. **MSP is underrated for OOD.** Maximum softmax probability consistently matches or exceeds learned methods on out-of-distribution datasets, with zero computational overhead.

3. **Self-consistency dominates when cost permits.** At 10× inference cost, self-consistency achieves the highest HLCC on DeepSeek-R1-14B (1.933) and strong AUROC across models.
4. **Post-hoc calibration is powerful.** Isotonic regression and Platt scaling substantially improve discrimination without any model retraining.

NormShift Case Study. As a case study in learned confidence, the NormShift head achieves near-perfect in-domain discrimination on Mistral-7B (AUROC = 0.999) and effective confidence-gated retry (32% rescue rate on TruthfulQA, zero damage). However, performance is architecture-dependent and degrades on OOD data, consistent with the general finding that learned methods trade generality for in-domain precision. The per-layer signal analysis reveals that different architectures encode confidence at different depths (L27 for Mistral-7B, L2 for Llama-3.1-8B), suggesting that normalisation dynamics carry architecture-specific self-knowledge.

Future Work. Two directions merit investigation. First, training with *unbounded* HLCC as an objective, where the confidence head outputs $c \in [0, \infty)$ rather than $[0, 1]$, could better exploit the incentive gradient for near-certain predictions. Second, domain-agnostic training strategies (e.g., meta-learning across datasets) could address the OOD transfer limitation of learned confidence heads.

A Full Calibration Scores by Method

This appendix reports the complete per-method calibration scores evaluated on TruthfulQA (300 examples). These results complement the summary comparisons in Section 6 by providing the full metric suite for each method and model.

A.1 Mistral-7B (FP16)

Table 14: Calibration scores for Mistral-7B on TruthfulQA (300 examples, FP16).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Phase 1: Logit-based							
Entropy	.467	.389	.393	.105	.602	.531	.159
Top-k	.467	.428	.422	.060	.602	.545	.194
MSP	.467	.353	.352	.214	.711	.706	.513
GNLL	.467	.353	.352	.214	.711	.706	.513
Phase 2: Generation-based							
Verbalized	.467	.502	.516	−0.075	.595	.557	.222
Self-consistency	.290	.422	.410	−0.254	.556	.338	.114
CoT confidence	.513	.463	.449	.124	.581	.601	.214
Semantic entropy	.300	.328	.354	−0.076	.518	.323	.054
LVU	.467	.122	.287	.412	.464	.453	−0.044
CoCoA	.277	.506	.456	−0.385	.585	.323	.115
Phase 2: Post-hoc calibration†							
Temp. scaling	.467	.515	.510	−0.090	.651	.641	.393
Isotonic†	.467	**.000**	.230	.485	.654	.627	.368
Platt†	.467	.050	.243	.462	.602	.531	.159
Learned: NormShift (combined)‡							
NormShift‡	.467	.030	**.025**	**.884**	**.999**	**.999**	**.999**

†Post-hoc methods fit and evaluate on the same data.
‡NormShift includes TruthfulQA in training data.

A.2 Qwen2.5-7B (FP16)

Table 15: Calibration scores for Qwen2.5-7B on TruthfulQA (300 examples, FP16).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Phase 1: Logit-based							
Entropy	.493	.449	.425	.125	.899	.895	.834
Top-k	.493	.476	.465	.050	.757	.769	.601
MSP	.493	.277	.310	.340	.657	.589	.228
GNLL	.493	.277	.310	.340	.657	.589	.228
Phase 2: Generation-based							
Verbalized	.493	.311	.404	.399	.384	.419	−0.253
Self-consistency	.683	.269	.268	**.825**	.626	.754	.254
CoT confidence	.657	.262	.277	.726	.637	.752	.312
Semantic entropy	.687	.261	.277	.824	.602	.739	.193
LVU	.493	.130	.285	.518	.554	.510	.037
CoCoA	.677	.266	.267	.801	.806	.904	.726
Phase 2: Post-hoc calibration†							
Temp. scaling	.493	.487	.487	.009	.652	.586	.235
Isotonic†	.493	**.000**	**.119**	.746	**.909**	**.912**	**.860**
Platt†	.493	.037	.130	.731	.899	.895	.834
Learned: NormShift (combined)‡							
NormShift‡	.493	.305	.339	.460	.633	.574	.211

†Post-hoc methods fit and evaluate on the same data.

‡NormShift includes TruthfulQA in training data.

A.3 Llama-3.1-8B (FP16)

Table 16: Calibration scores for Llama-3.1-8B on TruthfulQA (300 examples, FP16).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Phase 1: Logit-based							
Entropy	.393	.524	.491	−0.211	.809	.741	.653
Top-k	.393	.569	.548	−0.314	.746	.626	.481
MSP	.393	.295	.312	.144	.658	.527	.302
GNLL	.393	.295	.312	.144	.658	.527	.302
Phase 2: Generation-based							
Verbalized	.393	.421	.565	−0.198	.467	.371	−0.069
Self-consistency	.507	.242	.279	.440	.707	.716	.485
CoT confidence	.647	.213	.261	**.744**	.630	.741	.293
Semantic entropy	.507	.205	.253	.526	.691	.744	.527
LVU	.393	.120	.251	.314	.545	.459	.143
CoCoA	.523	.301	.309	.393	.722	**.777**	.577
Phase 2: Post-hoc calibration†							
Temp. scaling	.393	.566	.558	−0.333	.636	.485	.217
Isotonic†	.393	**.000**	**.161**	.481	**.828**	.751	**.668**
Platt†	.393	.075	.177	.469	.809	.741	.653
Learned: NormShift (combined)‡							
NormShift‡	.393	.235	.253	.324	.794	.713	.613

†Post-hoc methods fit and evaluate on the same data.

‡NormShift includes TruthfulQA in training data.

A.4 Qwen2.5-14B (8-bit)

Table 17: Calibration scores for Qwen2.5-14B on TruthfulQA (300 examples, 8-bit).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Phase 1: Logit-based							
Entropy	.620	.322	.311	.600	.901	.932	.845
Top-k	.620	.352	.342	.549	.799	.838	.624
MSP	.620	.240	.263	.694	.719	.796	.508
GNLL	.620	.240	.263	.694	.719	.796	.508
Phase 2: Generation-based							
Verbalized	.620	.229	.298	.731	.604	.707	.264
Self-consistency	.740	.178	.180	1.113	.747	.859	.490
CoT confidence	.767	.189	.197	1.114	.689	.875	.486
Semantic entropy	.747	.171	.177	**1.140**	.740	.876	.542
LVU	.620	.262	.328	.736	.531	.667	.137
CoCoA	.740	.183	.188	1.082	.846	.936	.773
Phase 2: Post-hoc calibration†							
Temp. scaling	.620	.366	.361	.516	.669	.762	.420
Isotonic†	.620	**.000**	**.112**	1.007	**.910**	**.941**	**.867**
Platt†	.620	.057	.122	.984	.901	.932	.845
Learned: NormShift (combined)‡							
NormShift‡	.620	.183	.201	.840	.786	.811	.558

†Post-hoc methods fit and evaluate on the same data.

‡NormShift includes TruthfulQA in training data.

A.5 DeepSeek-R1-14B (8-bit)

Table 18: Calibration scores for DeepSeek-R1-14B on TruthfulQA (300 examples, 8-bit).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
Phase 1: Logit-based							
Entropy	.497	.373	.364	.225	.839	.838	.731
Top-k	.497	.383	.374	.211	.742	.748	.560
MSP	.497	.234	.302	.365	.602	.552	.143
GNLL	.497	.234	.302	.365	.602	.552	.143
Phase 2: Generation-based							
Verbalized	.497	.347	.371	.256	.560	.542	.121
Self-consistency	.983	.013	**.012**	1.933	**.984**	**1.000**	**.985**
CoT confidence	1.000	.500	.250	1.500	.500	.500	.000
Semantic entropy	.990	.028	.017	**1.937**	.961	**1.000**	.960
LVU	.497	.256	.351	.517	.484	.504	.007
CoCoA	.987	.071	.015	1.884	.976	**1.000**	.975
Phase 2: Post-hoc calibration†							
Temp. scaling	.497	.477	.480	.029	.597	.603	.262
Isotonic†	.497	**.000**	.153	.691	.853	.856	.760
Platt†	.497	.045	.164	.665	.839	.838	.731
Learned: NormShift (combined)‡							
NormShift‡	.497	.112	.113	.819	.948	.933	.898

†Post-hoc methods fit and evaluate on the same data.

‡NormShift includes TruthfulQA in training data.

B Cross-Dataset Evaluation (50/50 Split)

Each training dataset (ARC-Easy, TruthfulQA, ARC-Challenge) is split 50/50: the first half trains the NormShift head, the second half serves as the in-domain test set. Out-of-distribution (OOD) datasets are never seen during training. When NormShift confidence falls below 0.4, a retry pass re-evaluates the question with a trick-question system prompt.

B.1 Mistral-7B

Table 19: Cross-dataset evaluation for Mistral-7B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.900	**.930**	.897	+0.033	1.751	1.651
Truthfulqa	in-dom	.620	**.978**	.763	+0.215	1.125	.186
Arc Challenge	in-dom	.760	**.919**	.793	+0.126	1.321	1.091
Commonsenseqa	OOD	.687	.713	**.764**	−0.051	.926	.919
Openbookqa	OOD	.747	.644	**.740**	−0.096	1.013	1.070
Sciq	OOD	.860	.801	**.834**	−0.033	1.479	1.534
Hellaswag	OOD	.560	.493	**.641**	−0.148	.312	.541
Mmlu	OOD	.413	**.649**	.638	+0.011	.332	.225

B.2 Qwen2.5-7B

Table 20: Cross-dataset evaluation for Qwen2.5-7B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.853	.716	**.887**	−0.171	1.337	1.846
Truthfulqa	in-dom	.447	**.706**	.620	+0.086	.358	.304
Arc Challenge	in-dom	.793	.636	**.892**	−0.256	1.152	1.438
Commonsenseqa	OOD	.733	.625	**.797**	−0.172	1.019	1.297
Openbookqa	OOD	.667	.641	**.812**	−0.171	.865	1.437
Sciq	OOD	.847	.523	**.832**	−0.309	1.214	1.686
Hellaswag	OOD	.567	.410	**.742**	−0.332	.402	.735
Mmlu	OOD	.493	.568	**.718**	−0.150	.390	.597

B.3 Llama-3.1-8B

Table 21: Cross-dataset evaluation for Llama-3.1-8B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.840	.483	**.901**	−0.418	.939	1.616
Truthfulqa	in-dom	.527	.634	**.647**	−0.013	.442	.101
Arc Challenge	in-dom	.667	.430	**.827**	−0.398	.675	1.164
Commonsenseqa	OOD	.520	.572	**.703**	−0.131	.524	1.136
Openbookqa	OOD	.613	.611	**.805**	−0.193	.708	1.157
Sciq	OOD	.867	.560	**.875**	−0.315	1.094	1.728
Hellaswag	OOD	.467	.596	**.732**	−0.136	.470	.710
Mmlu	OOD	.373	.558	**.643**	−0.086	.247	.101

B.4 Qwen2.5-14B

Table 22: Cross-dataset evaluation for Qwen2.5-14B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.993	.282	**1.000**	−0.718	1.454	1.972
Truthfulqa	in-dom	.593	**.763**	.726	+0.037	.647	.598
Arc Challenge	in-dom	.907	.737	**.839**	−0.102	1.301	1.748
Commonsenseqa	OOD	.867	.564	**.861**	−0.297	1.040	1.553
Openbookqa	OOD	.920	.684	**.878**	−0.194	1.367	1.720
Sciq	OOD	.953	.777	**.909**	−0.131	1.307	1.839
Hellaswag	OOD	.753	.626	**.724**	−0.098	.992	1.084
Mmlu	OOD	.520	.551	**.767**	−0.216	.548	.492

B.5 DeepSeek-R1-14B

Table 23: Cross-dataset evaluation for DeepSeek-R1-14B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.973	.779	**.949**	−0.170	1.904	1.878
Truthfulqa	in-dom	.467	**.963**	.641	+0.322	.775	.282
Arc Challenge	in-dom	.880	.659	**.872**	−0.213	1.576	1.553
Commonsenseqa	OOD	.793	.661	**.819**	−0.158	1.044	1.294
Openbookqa	OOD	.833	.687	**.838**	−0.151	1.352	1.439
Sciq	OOD	.953	.818	**.927**	−0.109	1.803	1.820
Hellaswag	OOD	.580	.622	**.790**	−0.168	.629	.796
Mmlu	OOD	.573	.664	**.751**	−0.087	.682	.650

B.6 Trick-Question Retry Summary

Table 24: NormShift-gated retry with trick-question prompt. When NormShift confidence < 0.4, the question is re-evaluated with a trick-question system prompt. Rescue = wrong→correct; Damage = correct→wrong.

Model	Dataset	Retried	Rescues	Damage	Net
Mistral-7B	Arc Easy	13	0	0	+0
	Truthfulqa	78	25	0	+25
	Arc Challenge	28	3	1	+2
	Commonsenseqa	33	2	3	−1
	Openbookqa	18	1	0	+1
	Sciq	8	0	2	−2
	Hellaswag	40	0	2	−2
	Mmlu	95	3	3	+0
Qwen2.5-7B	Arc Easy	64	0	16	−16
	Truthfulqa	72	3	9	−6
	Arc Challenge	71	2	9	−7
	Commonsenseqa	76	2	13	−11
	Openbookqa	75	0	26	−26
	Sciq	89	1	12	−11
	Hellaswag	82	5	14	−9
	Mmlu	82	2	13	−11
Llama-3.1-8B	Arc Easy	127	1	10	−9
	Truthfulqa	64	22	2	+20
	Arc Challenge	129	1	12	−11
	Commonsenseqa	122	4	40	−36
	Openbookqa	134	8	28	−20
	Sciq	107	1	10	−9
	Hellaswag	133	4	19	−15
	Mmlu	102	10	11	−1
Qwen2.5-14B	Arc Easy	78	0	0	+0
	Truthfulqa	20	1	1	+0
	Arc Challenge	83	1	5	−4
	Commonsenseqa	121	3	4	−1
	Openbookqa	71	0	1	−1
	Sciq	102	0	1	−1
	Hellaswag	89	2	2	+0
	Mmlu	144	7	12	−5
DeepSeek-R1-14B	Arc Easy	1	0	0	+0
	Truthfulqa	80	0	0	+0
	Arc Challenge	8	1	0	+1
	Commonsenseqa	93	1	1	+0
	Openbookqa	25	0	1	−1
	Sciq	11	0	0	+0
	Hellaswag	118	1	5	−4
	Mmlu	89	0	1	−1

C Training Data Composition

Table 25: Training and evaluation data for NormShift confidence heads. All five models use identical data splits.

Split	Dataset	Examples
Training (900 total)	ARC-Easy	300
	TruthfulQA	300
	ARC-Challenge	300
Validation	TruthfulQA	100
Calibration eval	TruthfulQA	300

Dataset overlap. TruthfulQA appears in three roles: 300 training examples, 100 validation examples (used for early stopping / checkpoint selection), and 300 calibration evaluation examples. While we do not verify whether the specific question instances overlap between these splits, the presence of TruthfulQA in the training set means that calibration results on TruthfulQA should be considered *in-distribution* performance.

The strong AUROC (0.999) of NormShift on Mistral-7B TruthfulQA versus its degradation on held-out datasets in Table 8 directly reflects this overlap. The 50/50 split evaluation (Tables 19 and 20) provides a cleaner test: NormShift still dominates in-domain for Mistral-7B (AUROC 0.978 on held-out TruthfulQA) but not for Qwen2.5-7B (AUROC 0.706), and degrades on OOD datasets for both models.

Checkpoint vs. calibration evaluation gap. A notable discrepancy exists between the checkpoint training metrics and calibration evaluation. For example, the Qwen2.5-7B combined head achieves ECE = 0.060 at checkpoint selection, but ECE = 0.305 when evaluated as a calibration baseline on TruthfulQA. The Llama-3.1-8B head shows a similar gap (checkpoint ECE = 0.040 vs. calibration eval ECE = 0.235). Only Mistral-7B shows close agreement (checkpoint ECE = 0.020 vs. calibration eval ECE = 0.030). This gap suggests that the head overfits to its training distribution on some architectures, and that checkpoint metrics alone are insufficient for assessing generalisation.

D Methodology Notes

Post-hoc calibration data leakage. The post-hoc calibration methods (temperature scaling, isotonic regression, Platt scaling) fit their calibration parameters on the same 300 TruthfulQA examples used for evaluation. No held-out calibration set is used. As a result:

- Isotonic regression achieves ECE = 0.000 by construction—it perfectly maps raw confidences to observed frequencies on the fitting data. This does not reflect generalisation ability.
- Platt scaling achieves ECE = 0.050, also partially an artefact of fitting on the evaluation set.
- Temperature scaling achieves ECE = 0.515, which is *worse* than uncalibrated MSP (0.353), because the learned temperature exacerbates overconfidence for this model.

These results should be interpreted as *in-sample calibration fits* rather than true generalisation performance. A proper evaluation would use a separate calibration holdout (e.g., 80/20 split of the 300 examples).

NormShift in-distribution advantage. The NormShift confidence head is trained using HLCC loss on data that includes TruthfulQA examples. On Mistral-7B, this produces near-perfect discrimination (AUROC $\approx$ 1.0, PRR $\approx$ 1.0), and the 50/50 split confirms strong in-domain generalisation (AUROC 0.978 on held-out TruthfulQA, 0.930 on ARC-Easy, 0.919 on ARC-Challenge). However, this strong performance does not transfer to all architectures: on Qwen2.5-7B, NormShift achieves only AUROC 0.706 on held-out TruthfulQA and loses to MSP on two of three in-domain datasets.

On out-of-distribution data, AUROC drops to 0.49–0.80 (Mistral-7B) and 0.41–0.64 (Qwen2.5-7B). For Qwen2.5-7B, PRR metrics are unreliable as a comparison since the retry mechanism changes the accuracy.

This does not invalidate the norm-shift signal itself—the per-layer standard deviation patterns do differ between correct and incorrect predictions even on out-of-distribution data (Table 10). Rather, it indicates that the *learned decision boundary* is both distribution-specific and architecture-dependent. Mistral-7B's normalisation dynamics appear to provide a particularly informative basis for confidence estimation, while other architectures may require different training strategies or larger training sets to achieve comparable transfer.

Implications for general-purpose confidence. The current NormShift head should not be used as a general-purpose confidence estimator for open-ended text generation without retraining on data representative of the target domain. For applications requiring robust confidence estimates across diverse inputs, logit-based methods (particularly MSP or entropy) remain more reliable due to their implicit adaptation to the model's output distribution.

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